To Run or Not to Run

Using Pre-Snap Behavior to Predict Play Types in the NFL

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1. Overview/ Introduction:

For our project we decided to participate in the National Football League's Big Data Bowl competition for 2025. The purpose of this competition is to use player and team behavior right before the play begins to gain insight into the resulting play. We realized that the most important information a defense can have is whether the offense intends to execute a run or pass play. Knowing this information would allow the defensive team to move their players around to optimize their chances of getting a tackle or otherwise disrupting the play. We therefore decided to develop statistical models that can predict what kind of play the offense will execute based on game characteristics, team characteristics, and player behavior before and during the snap which we will explore in greater detail later on.

2. Data Sets:

The NFL provided several useful data sets for us to draw from for our analysis.

-**games.csv**: This set contains all of the game data for the first 9 weeks of the 2022 season.

-**plays.csv**: This set contains all of the play by play data.

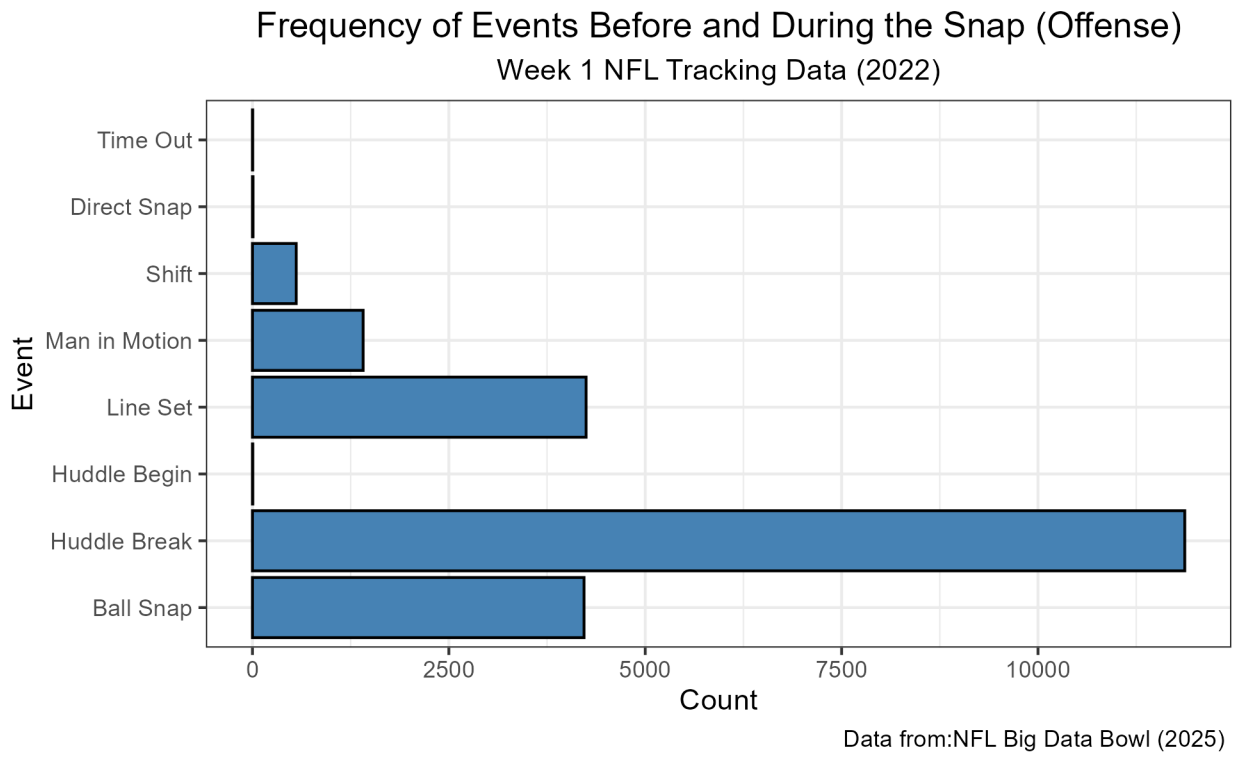
-**players.csv**: This set contains all of the individual performance data for NFL players.

-**player_play.csv**: This set contains all of the player level stats for each game and play.

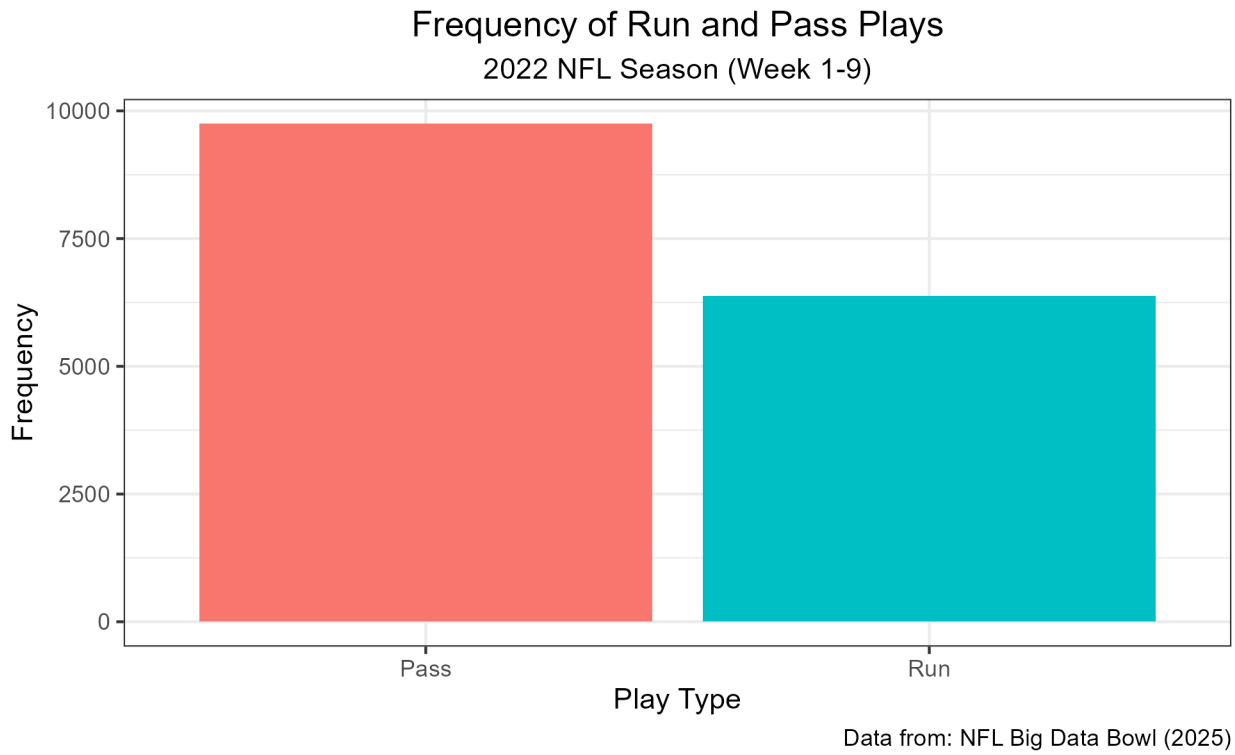
-**tracking_week_[week].csv**: This set contains all of the tracking data for players and the ball for each game and play in a given week.

3. Exploratory Data Analysis:

Since we were interested to see if things like player motion before or during the snap could be useful in predicting the play type we decided to look at a week of tracking data which contains key pre-snap events like when the offensive line is set (line_set), when the ball is snapped (ball_snap), and whether or not a player was in motion before the snap (man_in_motion) and plot the frequency of these events.



We were also interested to know if there would be any imbalance in our data (i.e. differing numbers of pass and run plays). To determine this we created indicator variables that classified any play with a dropback type of DESIGNED_RUN, and QB_SNEAK as a run play and any plays with a dropback type of TRADITIONAL, SCRAMBLE, or some form of ROLLOUT as pass plays. We then plotted the frequency of each type to visualize any imbalance.



4. Variables and Data Cleaning:

Using our background knowledge and drawing on our EDA we decided to include the following variables in our models.

Game Characteristics:

- **`game_id`**: Unique identifier for the game
- **`play_id`**: Unique identifier for a specific play
- **`possession_team`**: Identifies the offensive team
- **`week`**: Identifies which week of the season the game was in
- **`run_pass`**: Indicates if the play was a run or a pass (response variable)
- **`down`**: Identifies which down the offensive team was on for a particular play
- **`yards_to_go`**: The number of yards needed to get a first down
- **`yards_to_endzone`**: The number of yards to the opponent's endzone
- **`redzone`**: Indicates whether the offense is within 20 yards of the opponent's endzone
- **`sec_in_half`**: The number of seconds remaining on the play clock in the current half
- **`epa_lag`**: Serial correlation of expected points added

Team Characteristics:

- **`offense_formation`**: Categorical variable indicating what formation the offense uses
- **`receiver_alignment`**: Enumerated as 0x0, 1x0, 1x1, 2x0, 2x1, 2x2, 3x0, 3x1, 3x2 (text)
- **`curr_win_percentage`**: Win percentage for the current offensive team
- **`form_change`**: Uses shifts to determine if the offense changed formation before the ball was snapped

Player Characteristics:

- **`wr_motion`**: Indicates if a wide receiver moved after the line was set
- **`te_motion`**: Indicates if a tight end moved after the line was set
- **`rb_motion`**: Indicates if a running back moved after the line was set
- **`wr_atsnap`**: Indicates if a wide receiver was in motion at the time of the snap
- **`te_atsnap`**: Indicates if a tight end was in motion at the time of the snap
- **`rb_atsnap`**: Indicates if a running back was in motion at the time of the snap

5. Model Selection:

We were interested in predicting the type of play the offense will execute. We had a large number of predictors and the predictor space was further increased by our addition of several indicator variables for player motion. We therefore selected models that are good for binary classification and those that can handle larger predictor spaces without taking an excessive amount of time to run. We additionally wanted to try a non-parametric approach for comparison. These considerations lead us to choose the following models.

Logistic Regression (LASSO):

First we wanted to create a basic logistic regression model as "control" that we could compare our other more sophisticated models against. To accomplish this we used LASSO with 10-fold cross validation on the entire data set to perform variable selection. Once the optimal lambda value had been determined and the optimal LASSO model fit we look at the output to determine which of our variables to include in the logistic regression model. We then fit the logistic regression model on the entire data set in an attempt to identify any and all errors that would occurr later on when we tried to fit models for each team individually. Once this was complete we then fit a logistic regression model for each NFL team

Linear Support Vector Machine (SVM):

For the SVM Model we knew that the binary classification power would work well for our research question. There was a fairly high predictor space that we just left as and see if it could tackle them all before reduction would be required. This did take a while in practice and the k-fold cross validation had to be reduced from 10 folds to just 5 folds. Even with this reduction in k-folds the model still took quite a while to complete the training for all 32 teams. However as discussed later the prediction power of this method was retained and produced quite favorable results.

Random Forest:

KNN:

KNN was a non-parametric approach to the research question we tried. KNN can prove quite powerful when there is little overlap in the response compared to the feature space, which was our hope. However that did not prove to be the case and we ended abandoning this model all together. Part of the problem for KNN is the number of observations compared to the dimensionality of the problem. Predicting run or pass is a complicated endeavor and NFL offenses only get around 60 plays a game. Even when attempting to run on all 8-9 games for a team, that is only 540 games, or so, per team to train on. The model would run for some but failed to execute the prediction step for approximately half of the teams. We tried to make adjustments to the model to get predictions for every team but it was apparent KNN was not going to be a tool that we could use to describe the outcome effectively and so we decided to move on from this approach all together.

6. Model Evaluation:

To evaluate our models performance we decided to employ Matthews Correlation Coefficient (MCC). MCC which can be understood as the correlation coefficient between predicted and actual classifications can be expressed mathematically as follows:

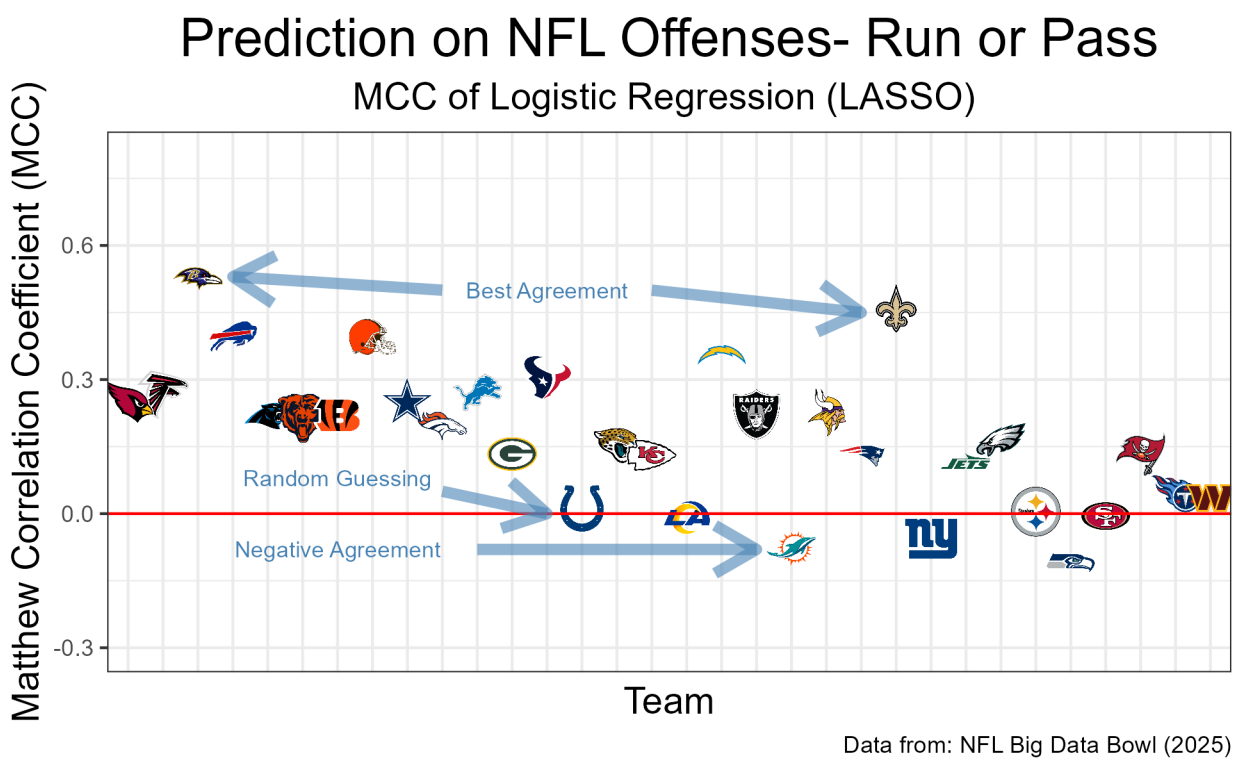
$$\text{MCC} = \frac{TP \times TN - FP \times FN}{\sqrt{(TP+FP)(TP+FN)(TN+FP)(TN+FN)}}$$

Here an MCC score of 1 indicates perfect agreement between predicted classes and actual classes, a score of -1 indicates complete disagreement between predicted and actual classes, and a score of 0 indicates that the predictions are no better than randomly guessing the class.

We can see that the numerator rewards true positives and true negatives while penalizing false positives and false negatives. The deonominator ensures that MCC scores are normalized between -1 to 1.

We chose to utilize MCC for several reasons. First, MCC is good at handling imbalanced data, which would be useful considering that pass plays are meaningfully more frequent than run plays. Second, it is comprehensive, considering all elements of the confusion matrices. Finally, we chose MCC because unlike precision and accuracy, MCC is unbiased.

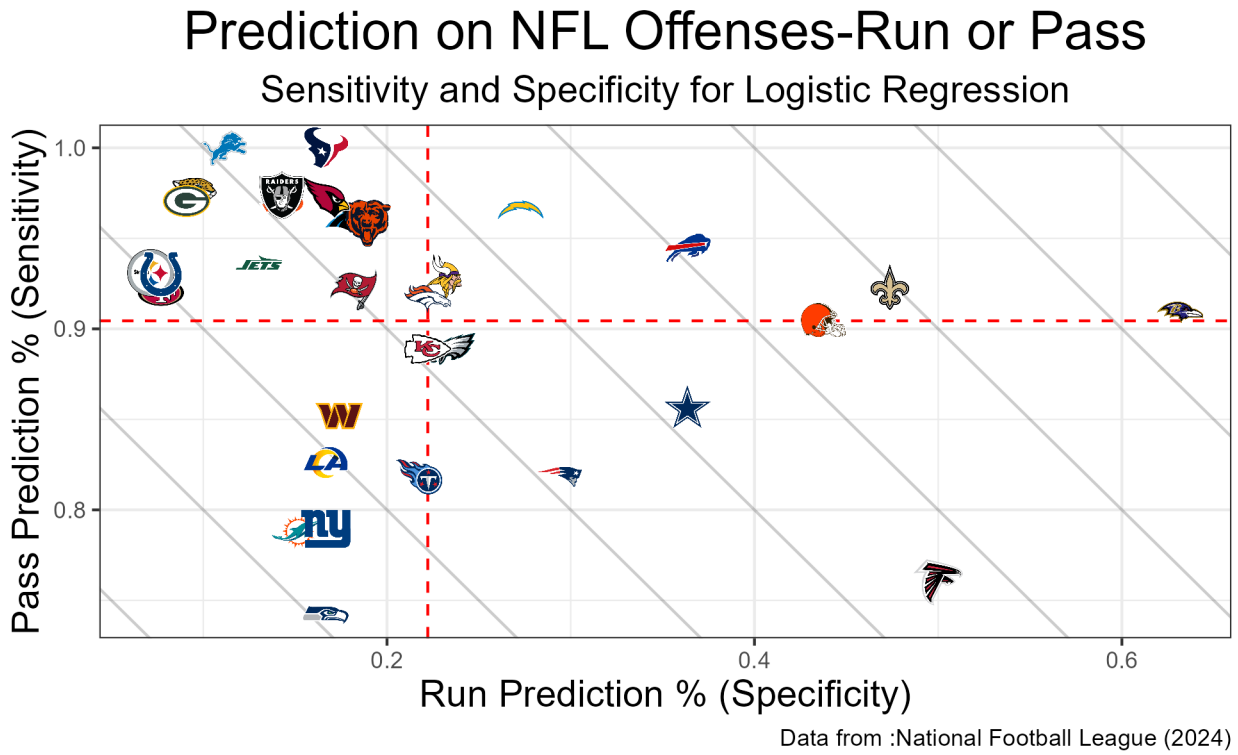
7. Results- Logistic Regression (LASSO):



In the above figure we can see the value of Matthews Correlation Coefficient for each teams logistic regression model. It is clear that most of the models showed modest levels of positive agreement between predictions and reality with the best agreement of $ 0.53$ occurring in the Baltimore Ravens model (top left), and the second best of 0.46 for the New Orleans Saints.

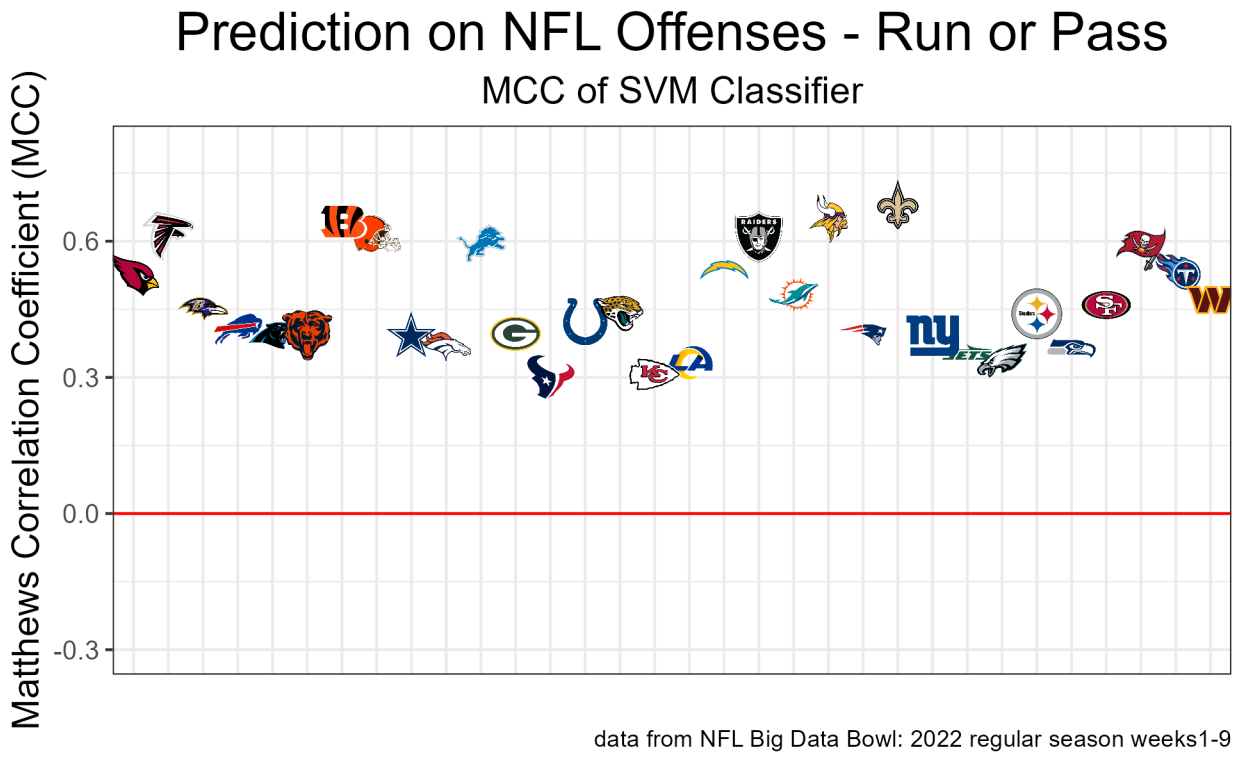
There were also several models with MCC scores of ≈ 0. These teams include the Indianapolis Colts, Los Angeles Rams, Pittsburgh Steelers, and the San Francisco 49'ers. The MCC score for these models indicate that their predictions for run or pass were no better than random guessing. This

Finally we can see that three of the models had negative MCC scores, namely the Miami Dolphins, The New York Giants, and the Seattle Seahawks. These MCC scores indicate that the models performed even worse than random guessing.

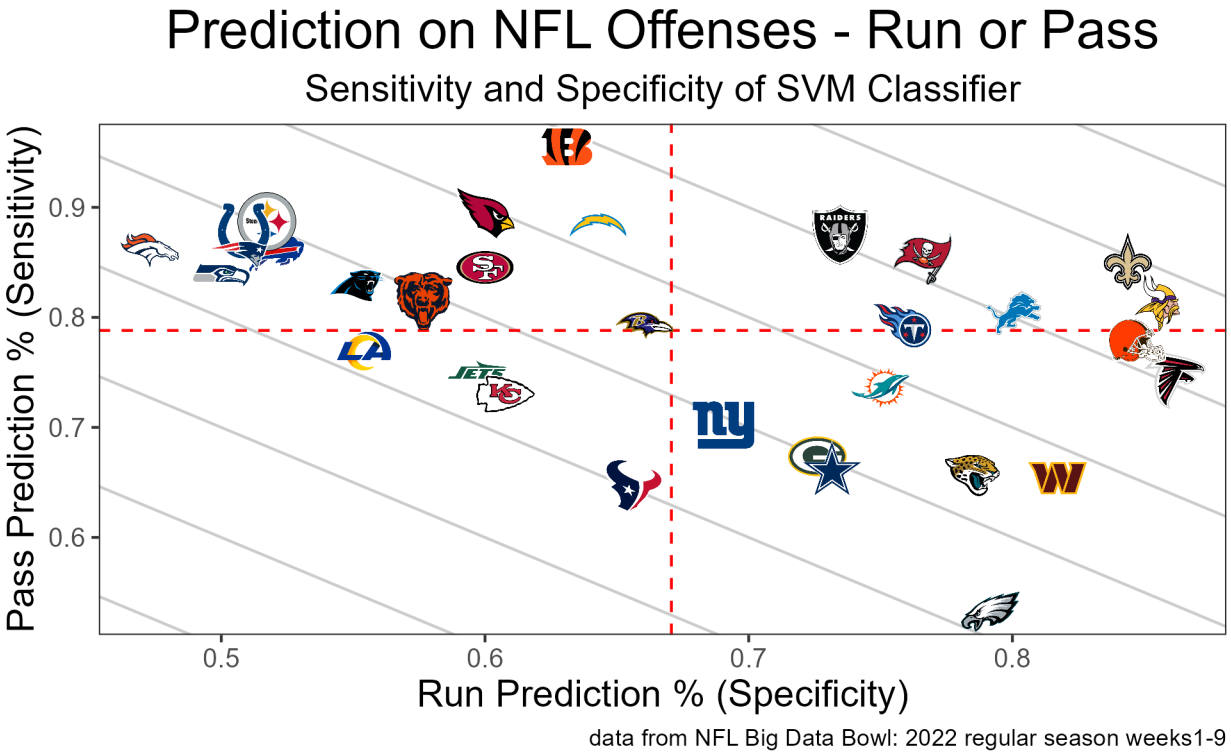


The above plot shows sensitivity (True Positive Rate) vs. specificity (True Negative rate) for each teams logistic regression model. The dotted red lines represent the mean sensitivity (≈ 0.905) and specificity (≈ 0.23) respectively. These means that the models were generally biased towards predicting passes, which is consistent with our earlier findings that pass plays are more common than run plays. The largest cluster of teams are located in the upper left quadrant where teams had above-average predictability for passing and below-average predictability for running. In the bottom left quadrant we there appears to be a lower bound on predictability for running. There appears to be much wider variation among teams with above average predictability for runs shown by the lesser amount of clustering on the right side of the graph.

8. Results- Linear SVM:



SVM performed well in terms of classification using the MCC. In the MCC plot for each team we see a band around 0.3 to 0.7 that each team lives in. Some are higher than others but all of them are above 0 indicating the SVM model performed quite well at predicting run or pass.



When we look at the breakdown between run and pass predictions in the Sensitivity vs Specificity plot, we can see a similar level of predictability for each team. As mentioned each team has a different proportion of run and pass plays, and this will influence the predictability. However, there is a line to the bottom left of the plot that no team falls below in. Almost a predictability minimum for the SVM model indicating a high level of performance, even for the teams lower in performance.

9. Results- Random Forest:

10. Model Comparison:

Team	Win %	Logistic MCC	SVM MCC	RF MCC
BUF	75.00	0.3957370	0.4055031	0.3684251
KC	75.00	0.1307441	0.3065196	0.3499566
MIA	66.67	-0.0769622	0.4819220	0.4747547
NYJ	66.67	0.1124527	0.3509050	0.3768834
BAL	66.67	0.5257862	0.4506045	0.5012716
TEN	62.50	0.0444554	0.5344837	0.4850150
LAC	62.50	0.3545986	0.5356372	0.5737523
CIN	55.56	0.2196874	0.6433699	0.6276743
NE	55.56	0.1284194	0.3929495	0.4404857
CLE	37.50	0.3952100	0.6162701	0.6599868
DEN	37.50	0.1964186	0.3676305	0.3594714
IND	33.33	0.0117307	0.4227842	0.4665414
JAX	33.33	0.1523123	0.4405956	0.5278097
PIT	25.00	0.0045305	0.4400517	0.4167629
LV	25.00	0.2217059	0.6096939	0.6566556
HOU	12.50	0.3042903	0.3004505	0.4057847

Team	Win %	Logistic MCC	SVM MCC	RF MCC
PHI	100.00	0.1646828	0.3371568	0.3115465
MIN	87.50	0.2265902	0.6498558	0.5664050
NYG	75.00	-0.0559717	0.3922067	0.4535598
DAL	75.00	0.2536560	0.3950867	0.3663374
SEA	66.67	-0.1097482	0.3631738	0.4378991
SF	50.00	-0.0054133	0.4595624	0.5536715
ATL	44.44	0.2664740	0.6113326	0.5531688
WAS	44.44	0.0350271	0.4703406	0.3803074
TB	44.44	0.1335909	0.5829315	0.6432400
LA	37.50	-0.0091264	0.3319766	0.3721414
NO	33.33	0.4596938	0.6807272	0.6228121
CHI	33.33	0.2134921	0.3927919	0.3639339
ARI	33.33	0.2523078	0.5281895	0.5378818
GB	33.33	0.1333855	0.3962696	0.3459265
DET	25.00	0.2721655	0.5959850	0.5460259
CAR	22.22	0.2183223	0.3924206	0.3809010

The above tables show the MCC values associated with each of our models by team. Here the models with the highest (best) MCC scores are colored in green while the lowest (worst) are colored in red. We can see that the logistic regression model performed poorly compared to the Random Forest and SVM with consistently lower MCC scores. The logistic model had the highest MCC score only once for the Baltimore Ravens team.

The performance of the random forest and SVM models were much more comparable with each having similar numbers of optimal MCC values. The SVM model was the optimal model for 16 of the 32 teams with most scores between 0.3 and 0.6. The random forest model was the optimal model for 15 of the 32 teams with most scores between 0.3 and 0.7. These findings suggest that the random forest and SVM models are equally likely to be the best choice, while the logistic regression model does not measure up.

11. Conclusion:

12. Citations:

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